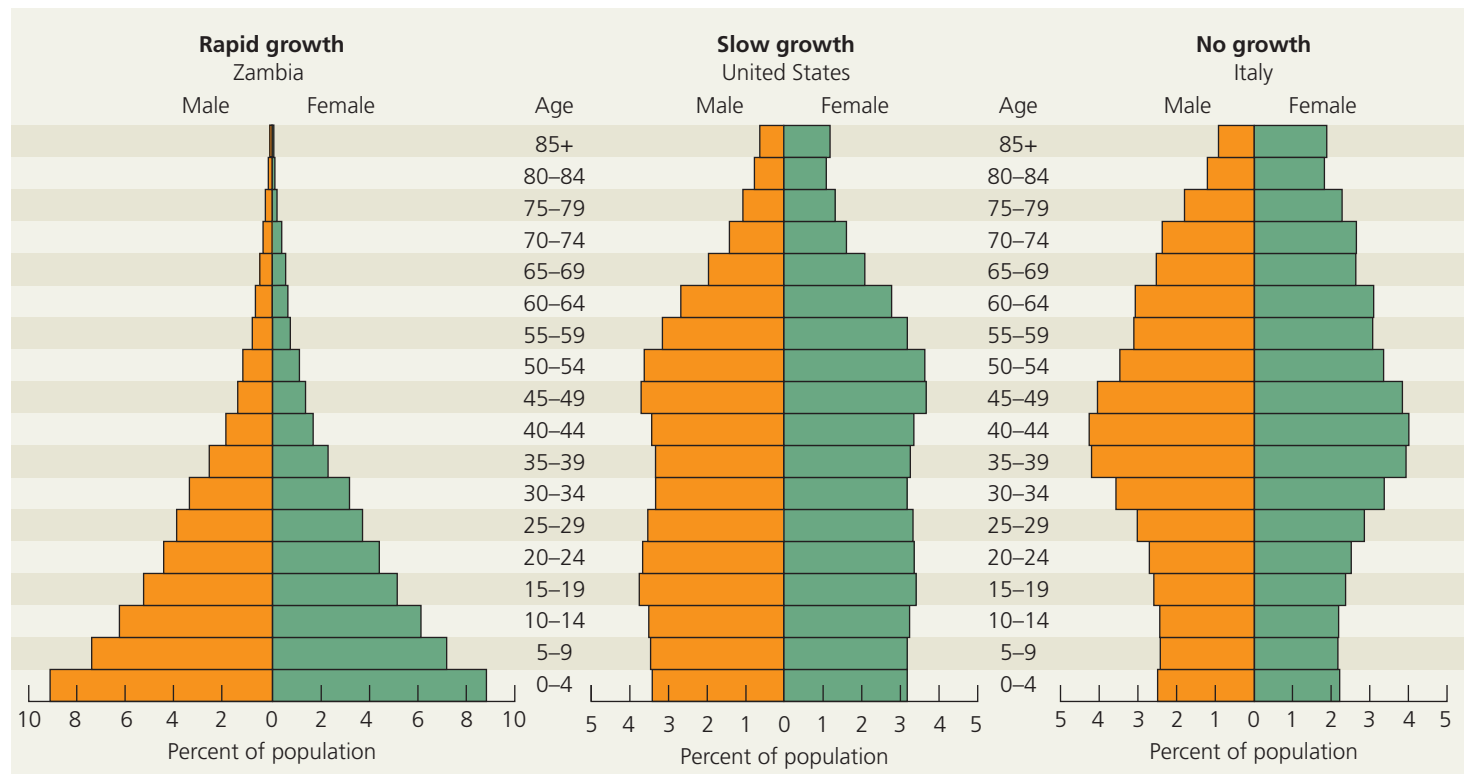


▼ **Figure 53.23 Age-structure pyramids for the human population of three countries.** The annual growth rate was approximately 2.8% in Zambia, 1.0% in the United States, and 0.0% in Italy. Note that Zambia's pyramid has a different scale for its x-axis from that of the other two pyramids. (Data as of 2010.)



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mortality is high, then parents may choose to have more children to ensure that some reach adulthood.

Although global life expectancy has been increasing since about 1950, it has recently dropped in a number of regions, including countries of the former Soviet Union and in sub-Saharan Africa. In these regions, social upheaval, decaying infrastructure, and infectious diseases such as AIDS and tuberculosis are reducing life expectancy.

Global Carrying Capacity

No ecological question is more important than the future size of the human population. As noted earlier, population ecologists project a global population of about 9.8 billion people in 2050. That means that over 2 billion people will be added to the population in the next four decades because of the momentum of population growth. But just how many humans can the biosphere support? Will the world be overpopulated in 2050? Is it *already* overpopulated?

Estimates of Carrying Capacity

For over three centuries, scientists have attempted to estimate the human carrying capacity of Earth. The first known

estimate, 13.4 billion people, was made in 1679 by Anton van Leeuwenhoek (a Dutch scientist who also discovered protists). Since then, estimates have varied from less than 1 billion to more than 1,000 billion (1 trillion).

Carrying capacity is difficult to estimate, and scientists use different methods to produce their estimates. Some researchers use curves like that produced by the logistic equation (see Figure 53.9) to predict the future maximum of the human population. Others generalize from existing “maximum” population density and multiply this number by the area of habitable land. Still others base their estimates on a single limiting factor, such as food, and consider variables such as the amount of farmland, the average yield of crops, the prevalent diet—vegetarian or meat based—and the number of calories needed per person per day.

Limits on Human Population Size

A more comprehensive approach to estimating the carrying capacity of Earth is to recognize that humans have multiple constraints: We need food, water, fuel, building materials, and other resources, such as clothing and transportation. The **ecological footprint** concept summarizes the aggregate land and water area required by each person, city, or nation